Communicating Your Work



a tutorial

AC297r

March 25, 2015

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Executive Director
Council for the Advancement of Science Writing
Institute for Applied Computational Science 2010–13
Harvard Initiative in Innovative Computing 2008–10

Editor, *American Scientist* magazine 1992–2008

today's topic

using words, pictures and symbols for elegant,clear communication to diverse audiences

agenda

general advice
presentation tips
an exercise
writing tips

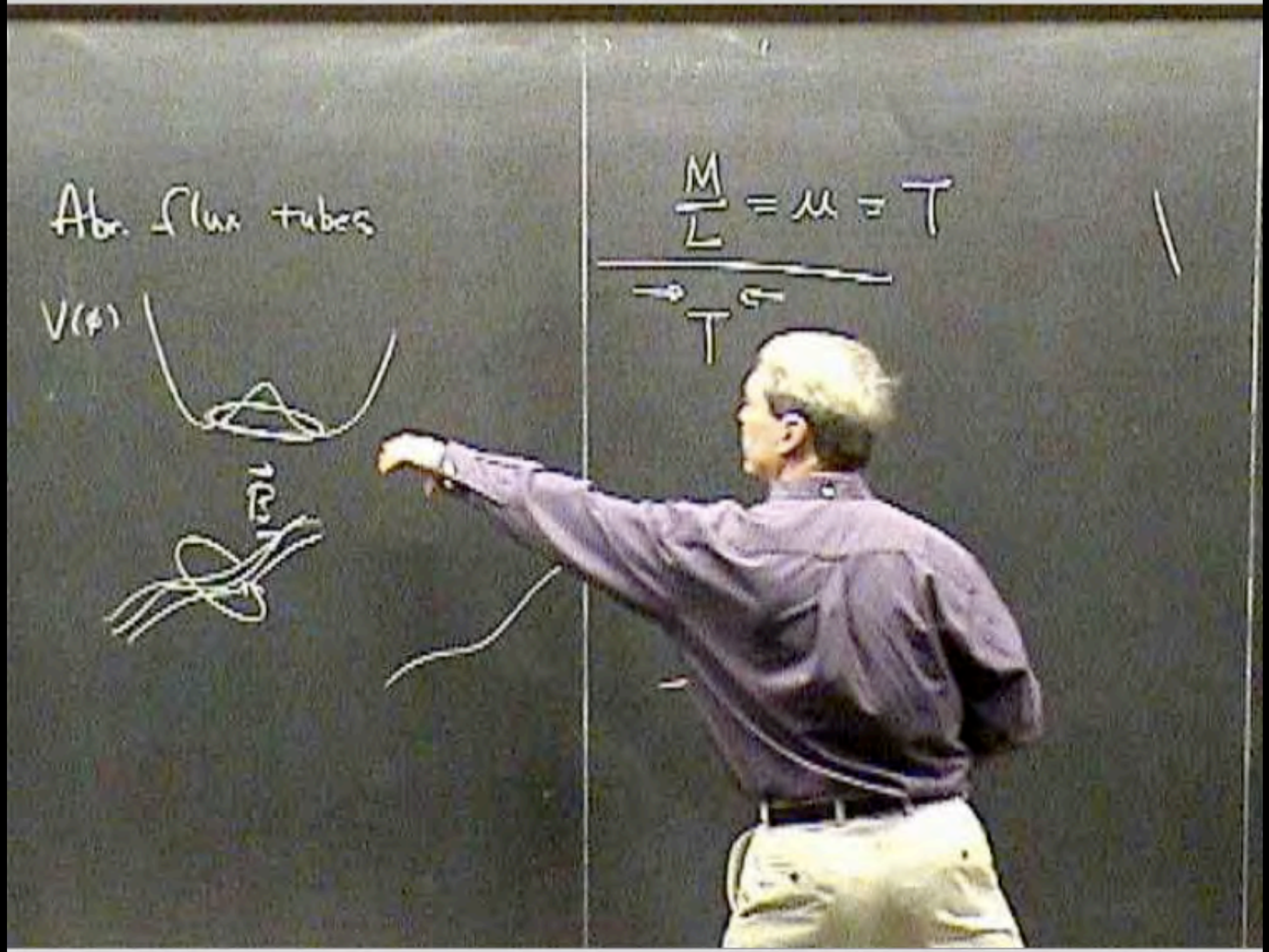
less is more


六玉河之内
紀伊高野


六玉河之内
陸奥野田

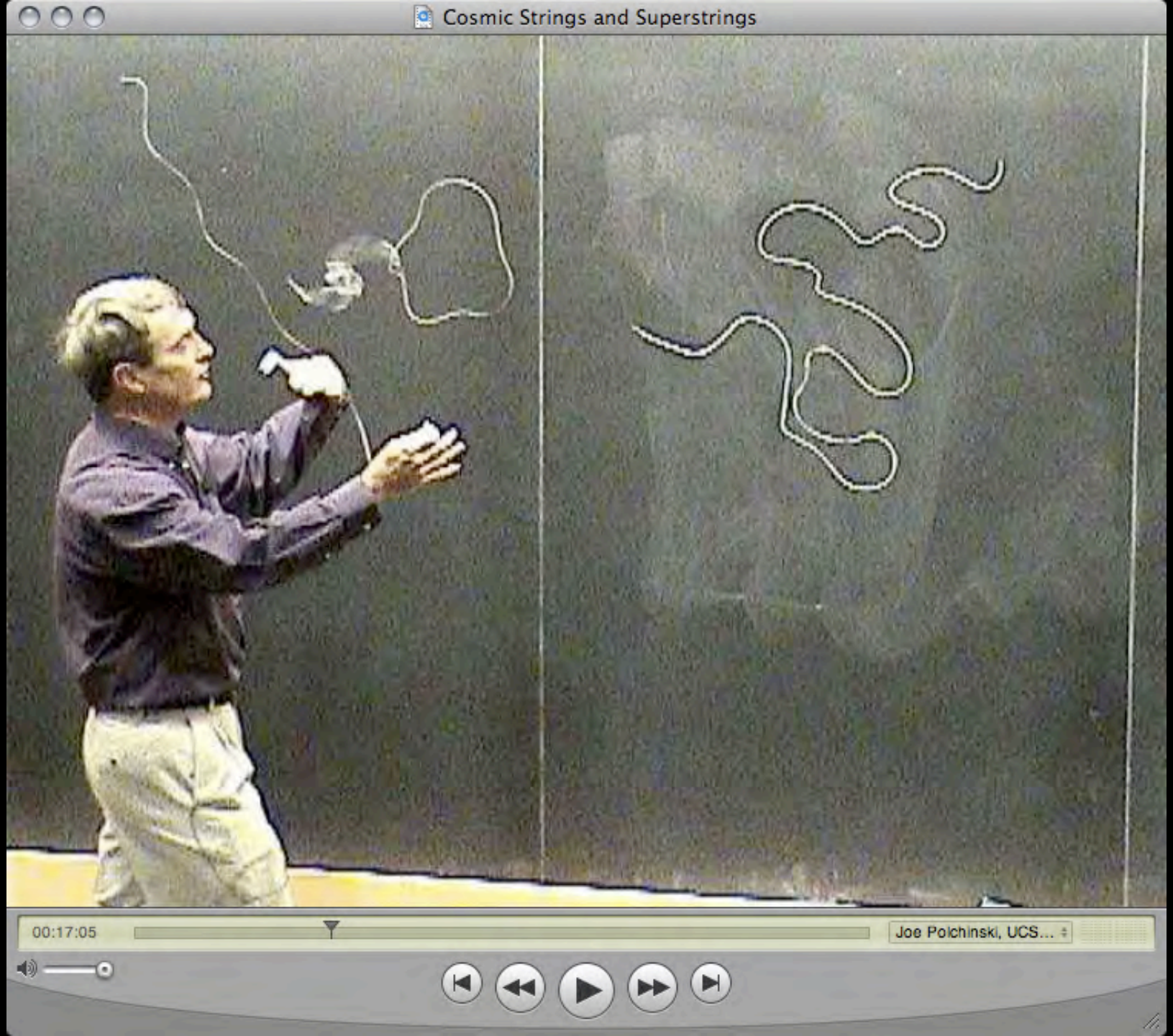
the scientist

Cosmic Strings and Superstrings

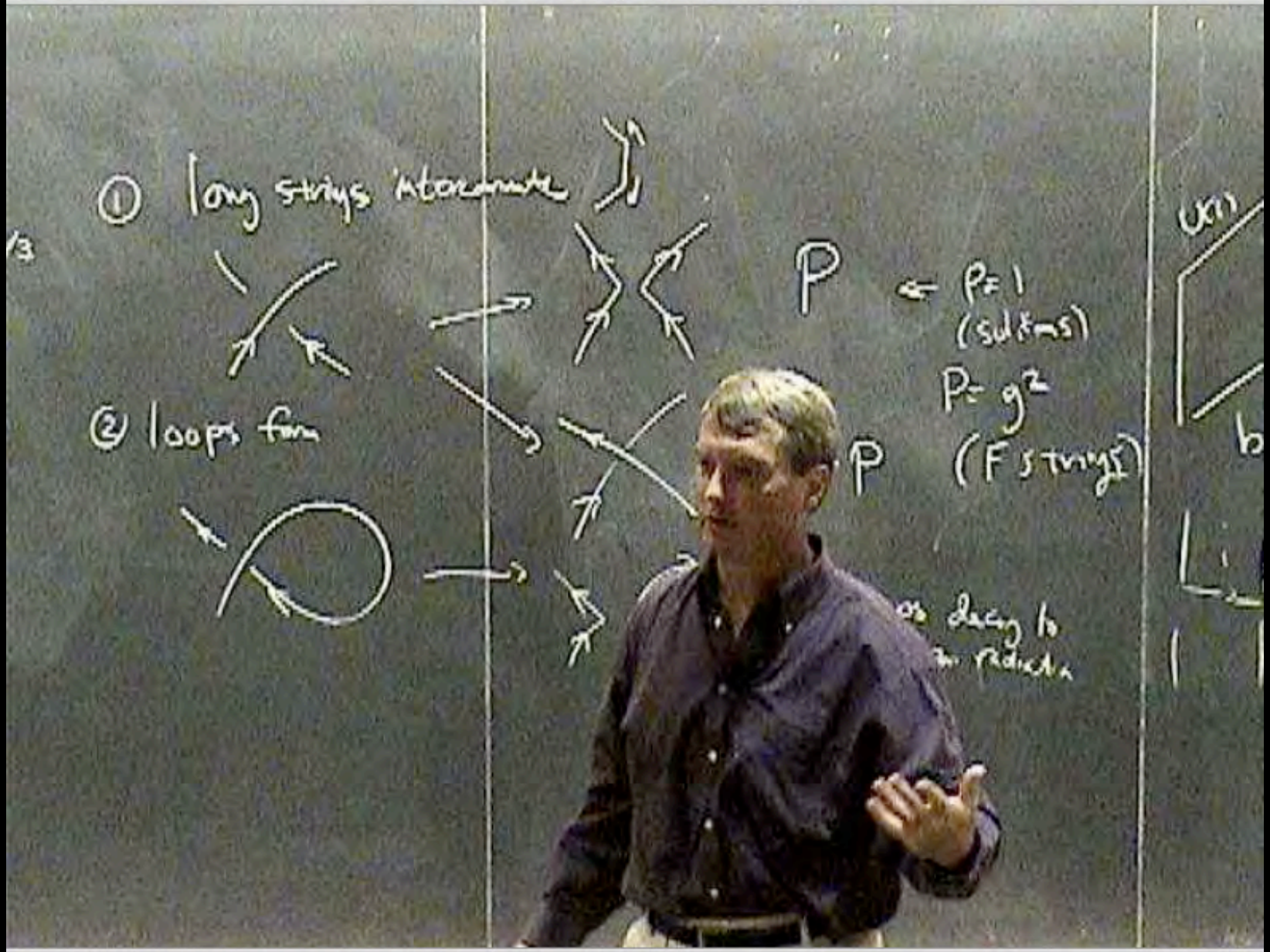


00:04:51

Joe Polchinski, UCS...


Cosmic Strings and Superstrings
00:17:05
Joe Polchinski, UCS...

Cosmic Strings and Superstrings

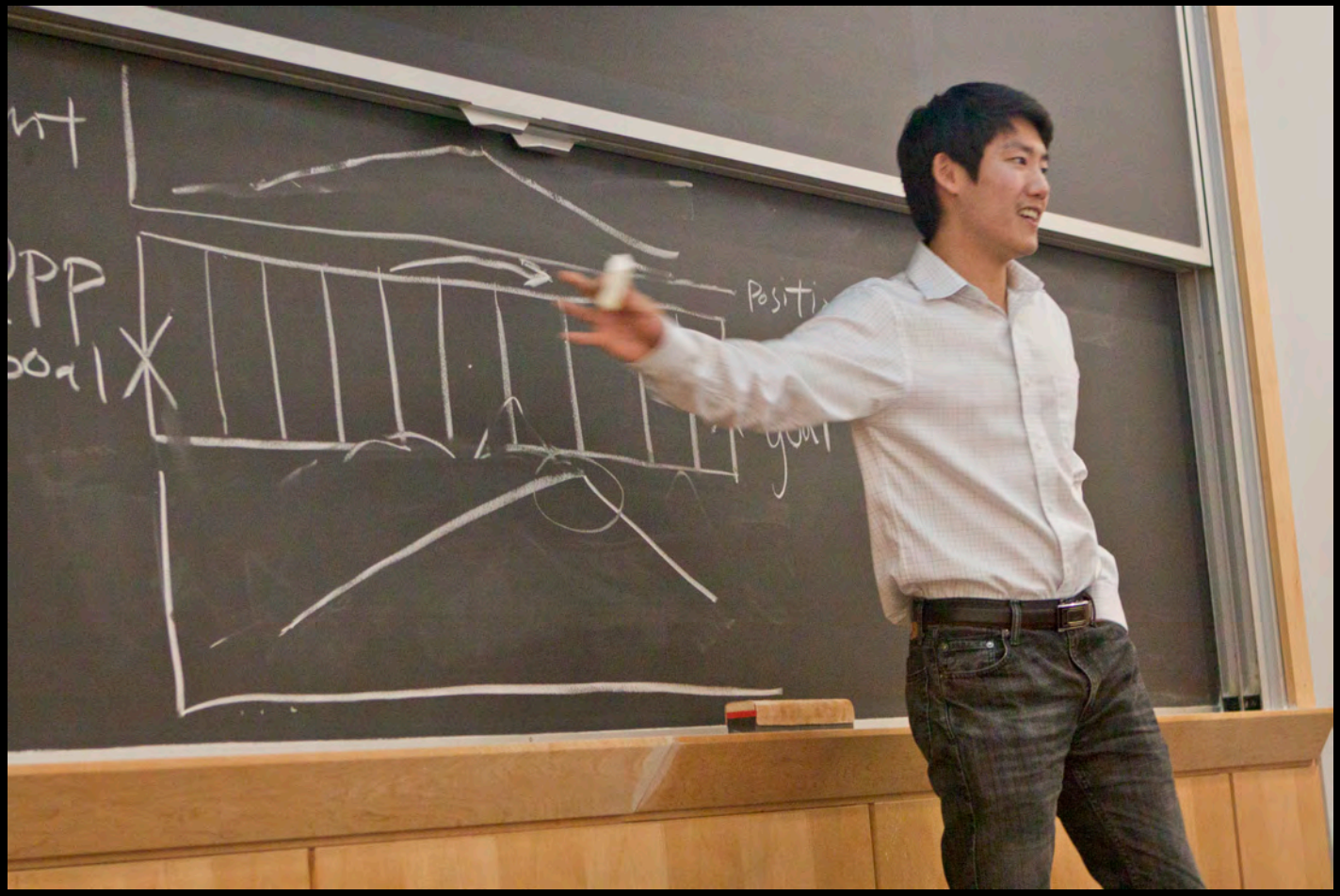


00:22:33

Joe Polchinski, UCS...

... tells stories
as humans have done for millennia

with words, symbols and pictures



how can you tell your story?

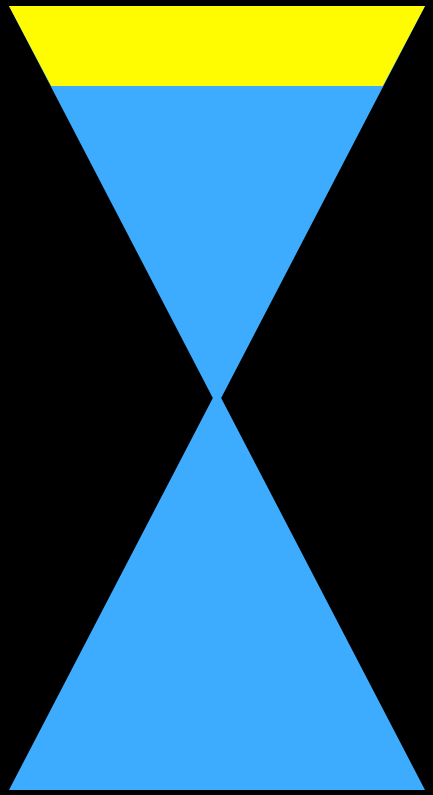
agenda

general advice
presentation tips
an exercise
writing tips

project presentations are stories

"I'm going to tell you about a problem and how I solved it."

story elements



1. motivation
description
approach
learning
result

story elements

1. motivation

why you (the listener) should care about what I'm going to say

Billions and Billions of Molecules

Exploring Chemical Space

http://aspuru.chem.harvard.edu
Twitter: **A_Aspuru_Guzik**
aspuru@chemistry.harvard.edu

Alán Aspuru-Guzik
Professor of Chemistry and Chemical Biology
Harvard University

"big picture"

what's in it for you

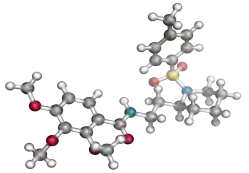
what's in it for the world

Some of the challenges of the 21st century

Clean Energy

Advanced drugs

Water purification

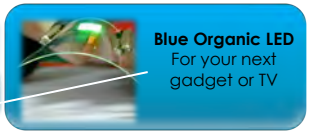


Molecules and materials

Long-lasting blue organic LED

The Harvard Clean Energy Project
Generating renewable energy





Blue Organic LED
For your next gadget or TV

Organic flow batteries
Storing renewable energy





Origins of life
How life may have come about?

Why OLEDs

Organic LED can be used to make thinner, foldable displays, with higher contrast ratios, since they do not require a backlight.

Display applications market expected to be $160 billion by 2017

Samsung Galaxy displays use OLED.

Iphone 6, Amazon fire, and LG G3 are LED-backlit IPS LCD.



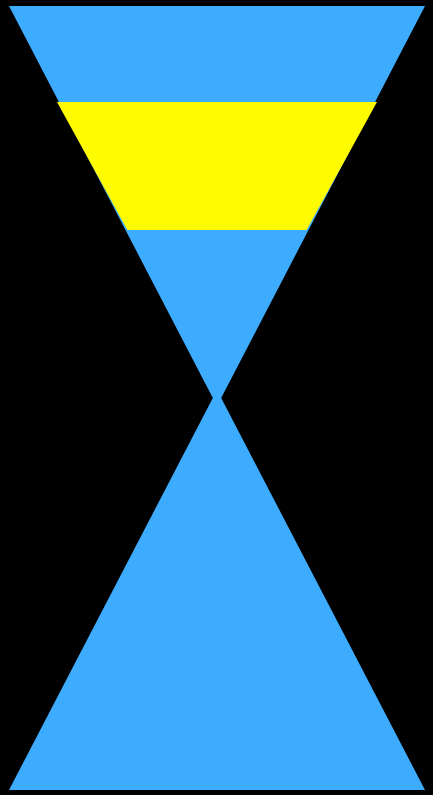
what motivates this group of listeners?

customize

- motivation
- technical level
- content

to each audience

story elements



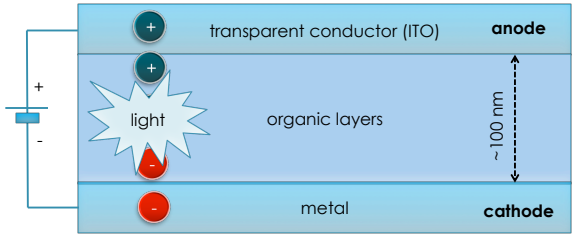
motivation
2. description
approach
learning
result

story elements

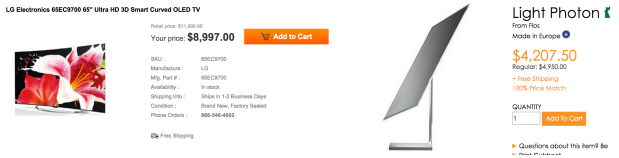
2. description

details of the problem,
and how you broke it down

How do OLEDs work?



Why doesn't everybody use OLED?



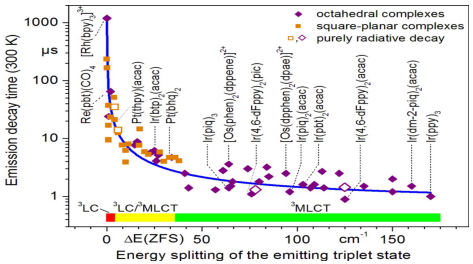
The blue phosphorescent molecules degrade very quickly!

technical background

practical interest

previous work

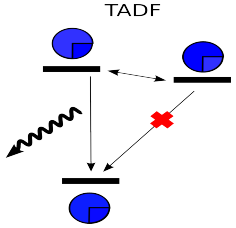
solution to be evaluated



Phosphorescence

Phosphorescent metal-organic OLEDs may have reached a physical limit.

Thermally activated delayed fluorescence

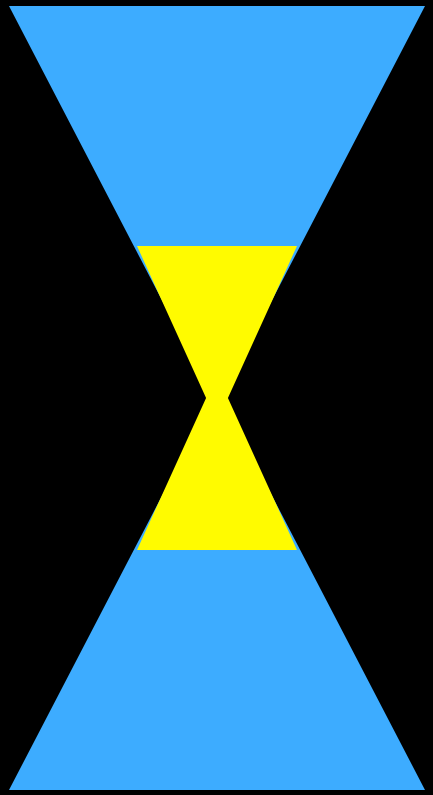


- S1-T1 splitting is low.
- Thermal agitation results in *Reverse* ISC.
- Can recover triplet excitons.
- 300% increase in efficiency.

$$r \propto \frac{H_{SO}}{\Delta_{ST}}$$

Pioneered by C. Adachi (Kyushu University)

story elements



motivation
description
3. approach
learning
result

story elements

3. approach

what you did, and why

this is the most important part of your story

- outline to identify key points
- strip out nonessential detail
- be clear

the presentation is **not** the paper

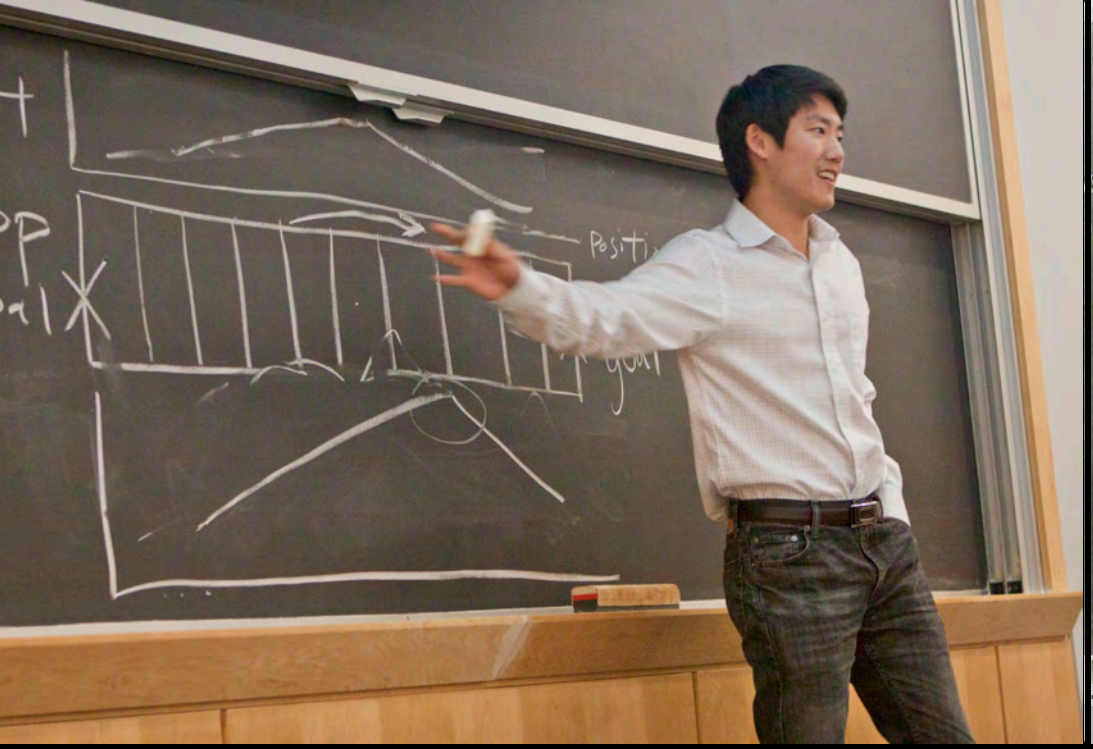
- do not try to answer all questions

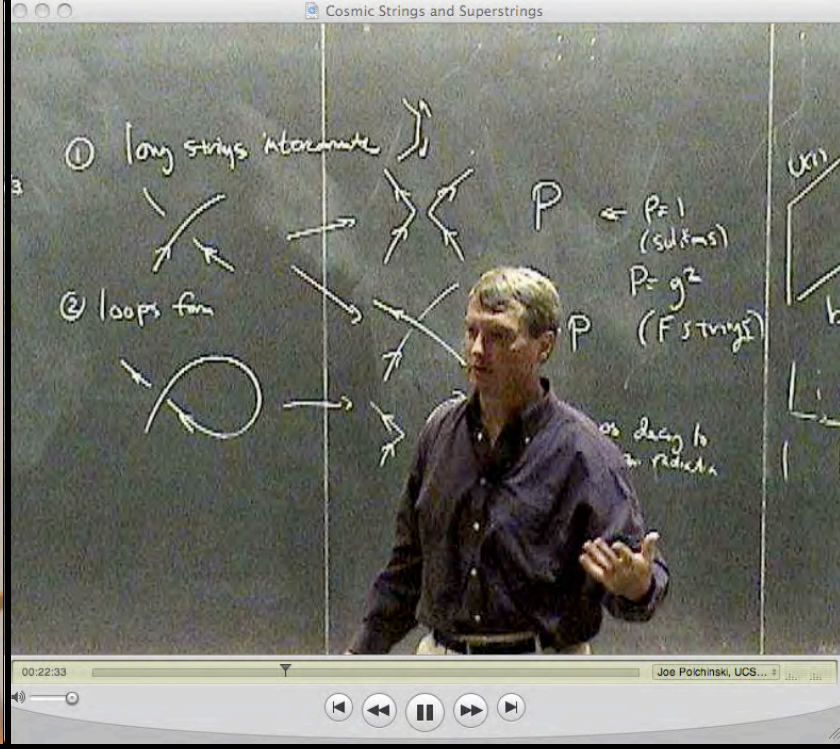
giving your talk

- don't rush
- look at listeners and watch for attention cues

human short-term memory holds 3–9 items at once





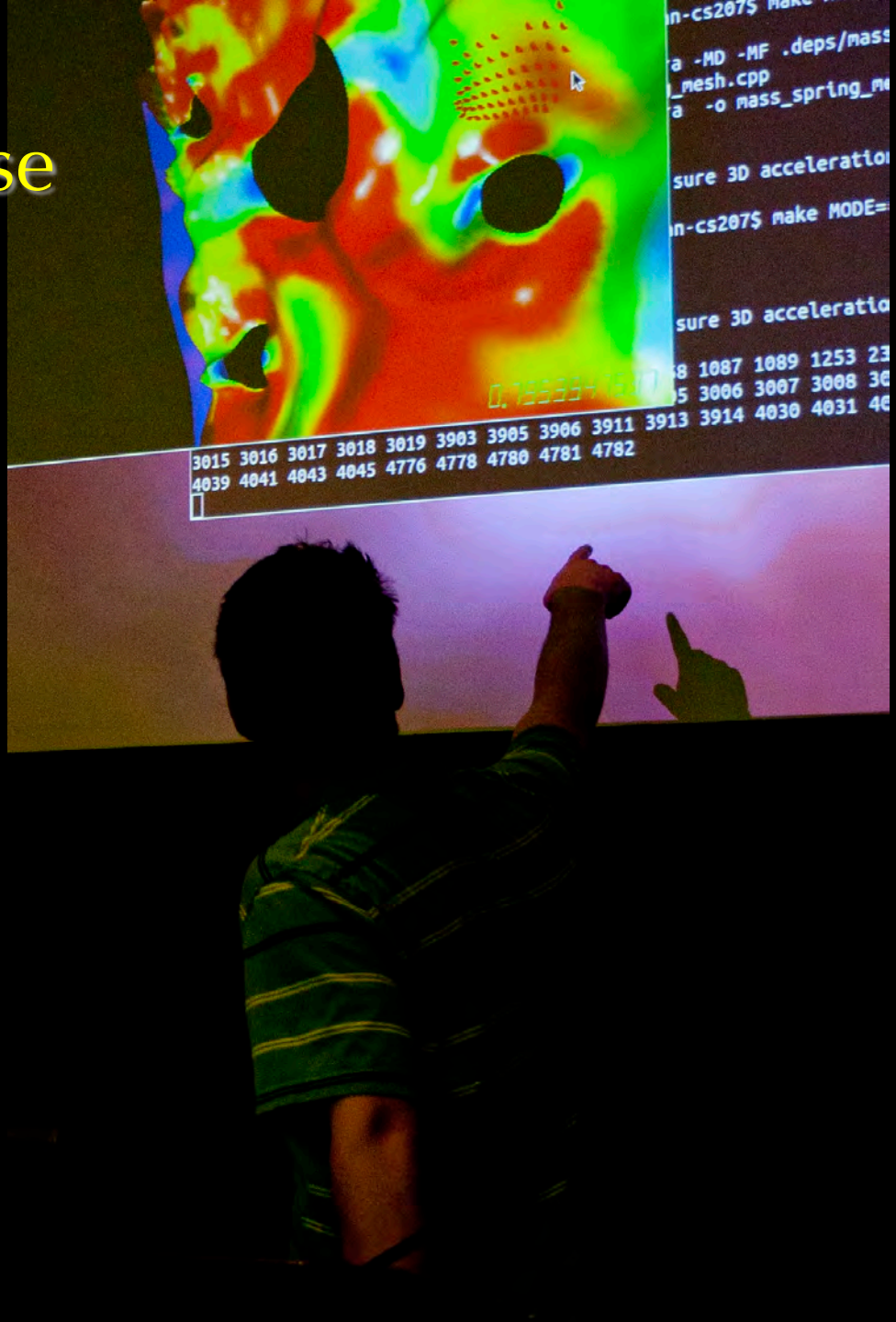


- engage with the audience
- speak with poise, confidence and dignity

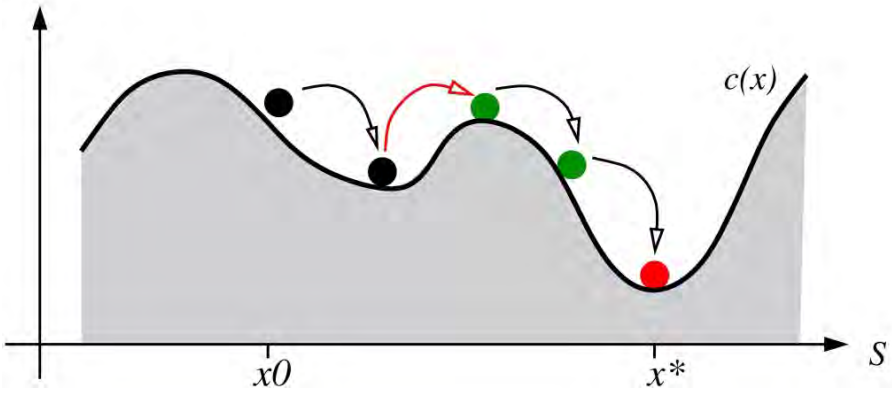
your slides

design slides for a diverse audience

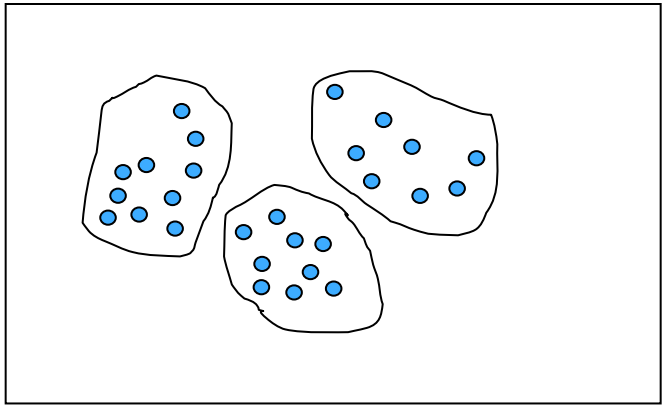
- use clear graphics
- didactic diagrams
- trim word count
- trim picture count
- few equations, well explained
- remove distractions, chartjunk



cartoon graphics enable the eye to aid understanding



simulated annealing



cluster analysis

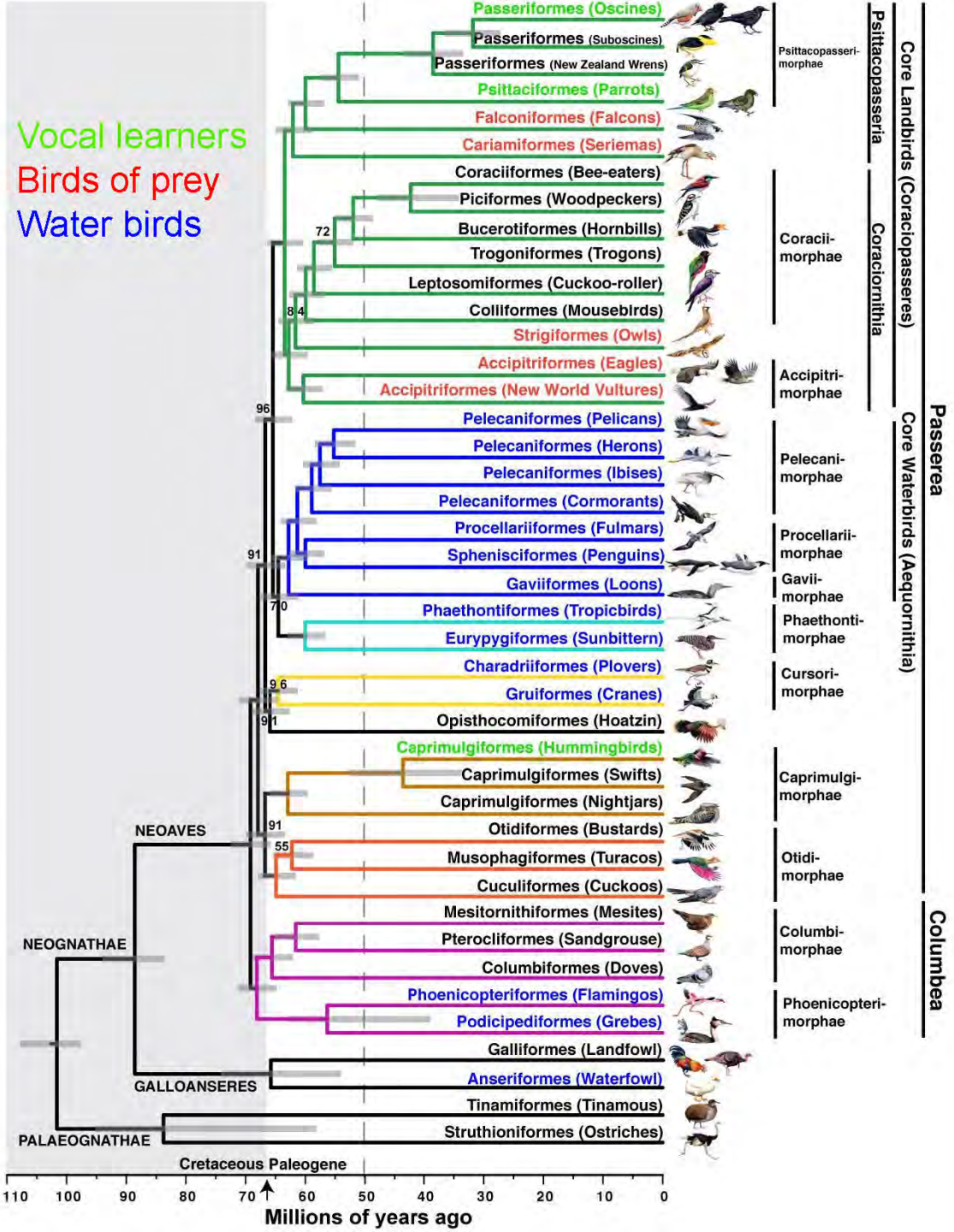
the presentation is **not** the paper

- do not put large chunks of story into a single slide
- assume each slide will be viewed for less than 60 seconds

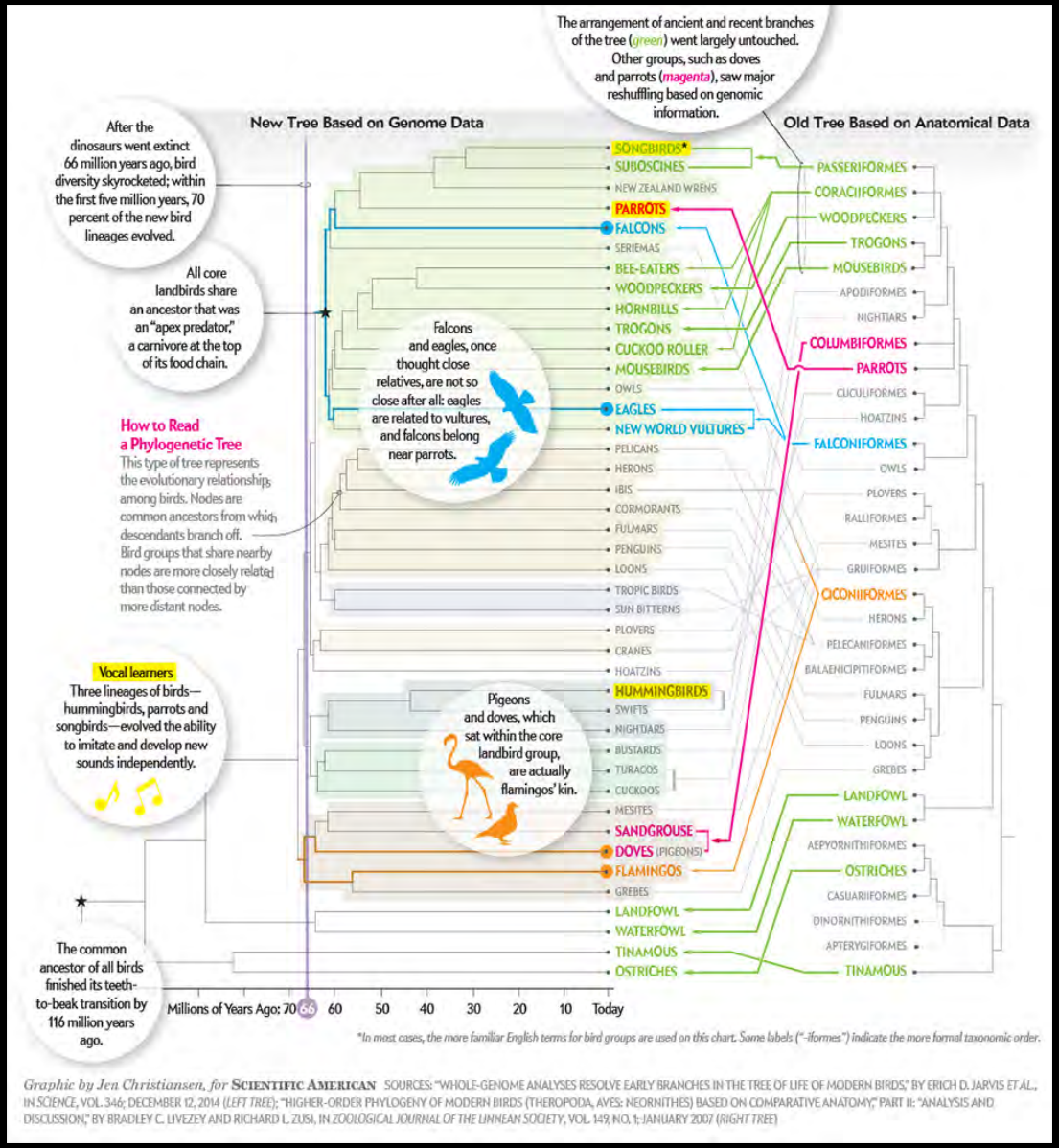
Our new genome scale tree

- Whole genome tree contradicts morphology-based and mitochondrial trees
- 2 to 4 independent gains of vocal learning, predatory, and water traits.
- Common ancestor of core land birds was a predator
- Rapid radiation of Neoves starting 66 MYA, as opposed to 80-100 MYA
- All orders appeared by 50 MYA.

Jarvis et al (in press)

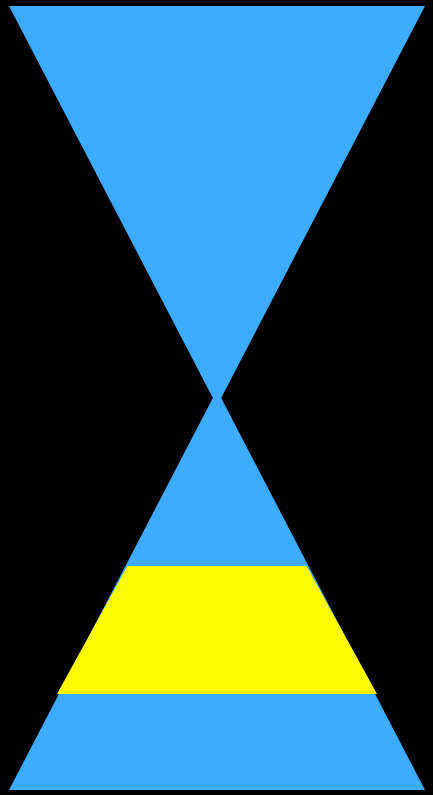


Scientific American infographic



Graphic by Jen Christiansen, for SCIENTIFIC AMERICAN SOURCES: "WHOLE-GENOME ANALYSES RESOLVE EARLY BRANCHES IN THE TREE OF LIFE OF MODERN BIRDS," BY ERICH D. JARVIS *ET AL.*, IN *SCIENCE*, VOL. 346; DECEMBER 12, 2014 (*LEFT TREE*); "HIGHER-ORDER PHYLOGENY OF MODERN BIRDS (THEROPODA, AVES: NEORNITHES) BASED ON COMPARATIVE ANATOMY," PART II: "ANALYSIS AND DISCUSSION," BY BRADLEY C. LIVEZEY AND RICHARD L. ZUSI, IN *ZOOLOGICAL JOURNAL OF THE LINNEAN SOCIETY*, VOL. 149, NO. 1; JANUARY 2007 (*RIGHT TREE*)

story elements



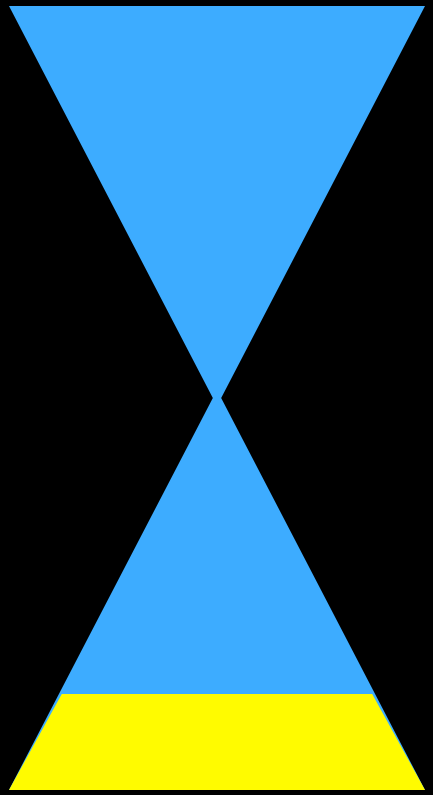
motivation
description
approach
4. learning
result

story elements

4. learning

insights, what **didn't** work

story elements



motivation
description
approach
learning
5. result

story elements

5. result

conclusion, discussion, implications

acknowledge supporters and teammates



References

Clean Energy Project
J. Phys. Chem. Lett. 2, 2241 (2011)
Energy Environ. Sci. 4, 4849 (2011)
Energy Environ Sci 7, 698 (2014)

Organic Flow Battery
Nature 505, 195 (2014)

Origins of life
J Comp
Theo Chem 10, 2097 (2014)

Organic electronics
Nat. Comm. 2, 437 (2011)
Nature 480, 504 (2011)

DOE, NSF, Samsung, Sloan Foundation
Camille and Henry Dreyfus Foundation

thank the audience for their attention

storyteller's tricks

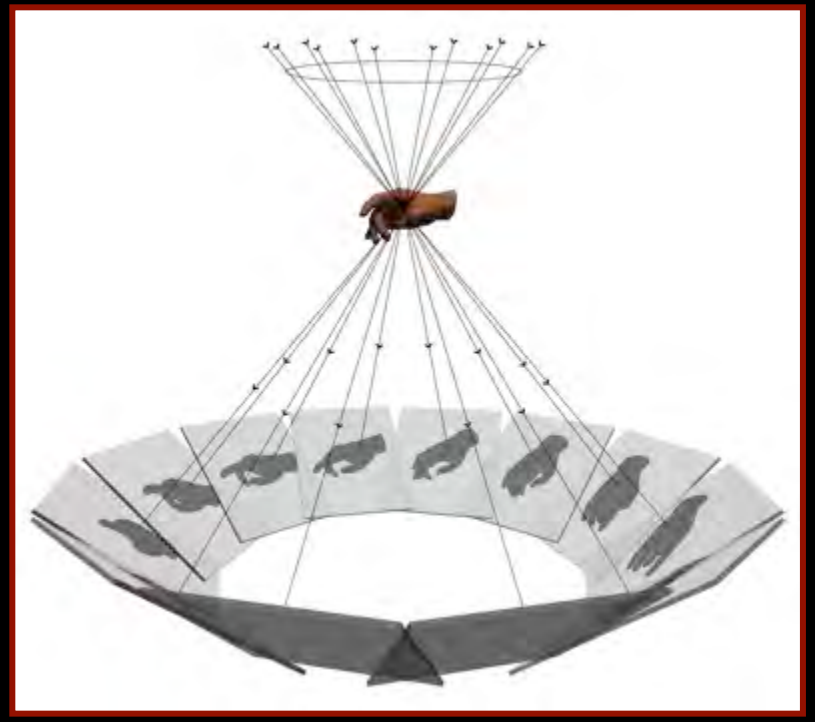
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for written, visual and spoken presentation

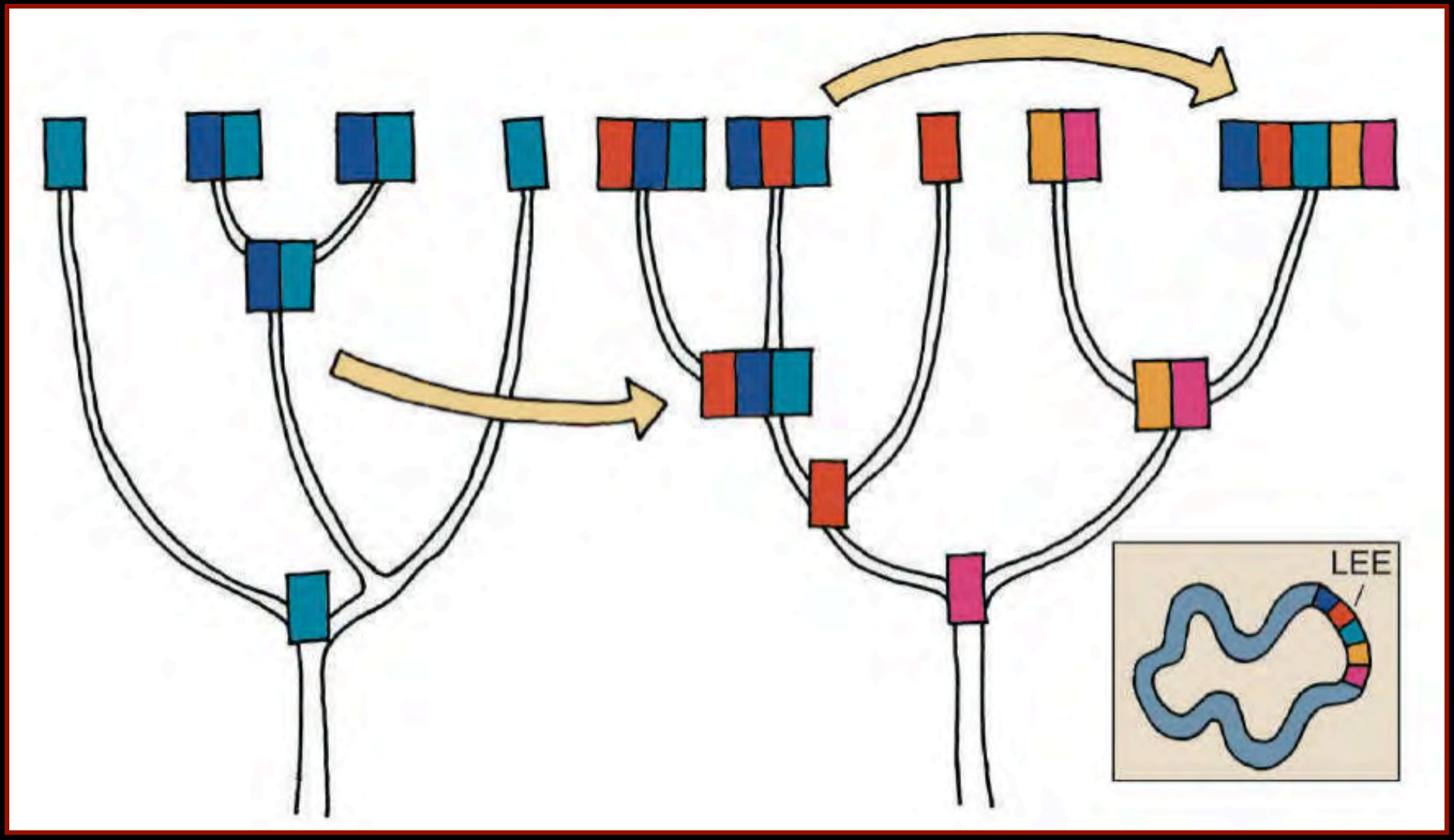
1. analogy



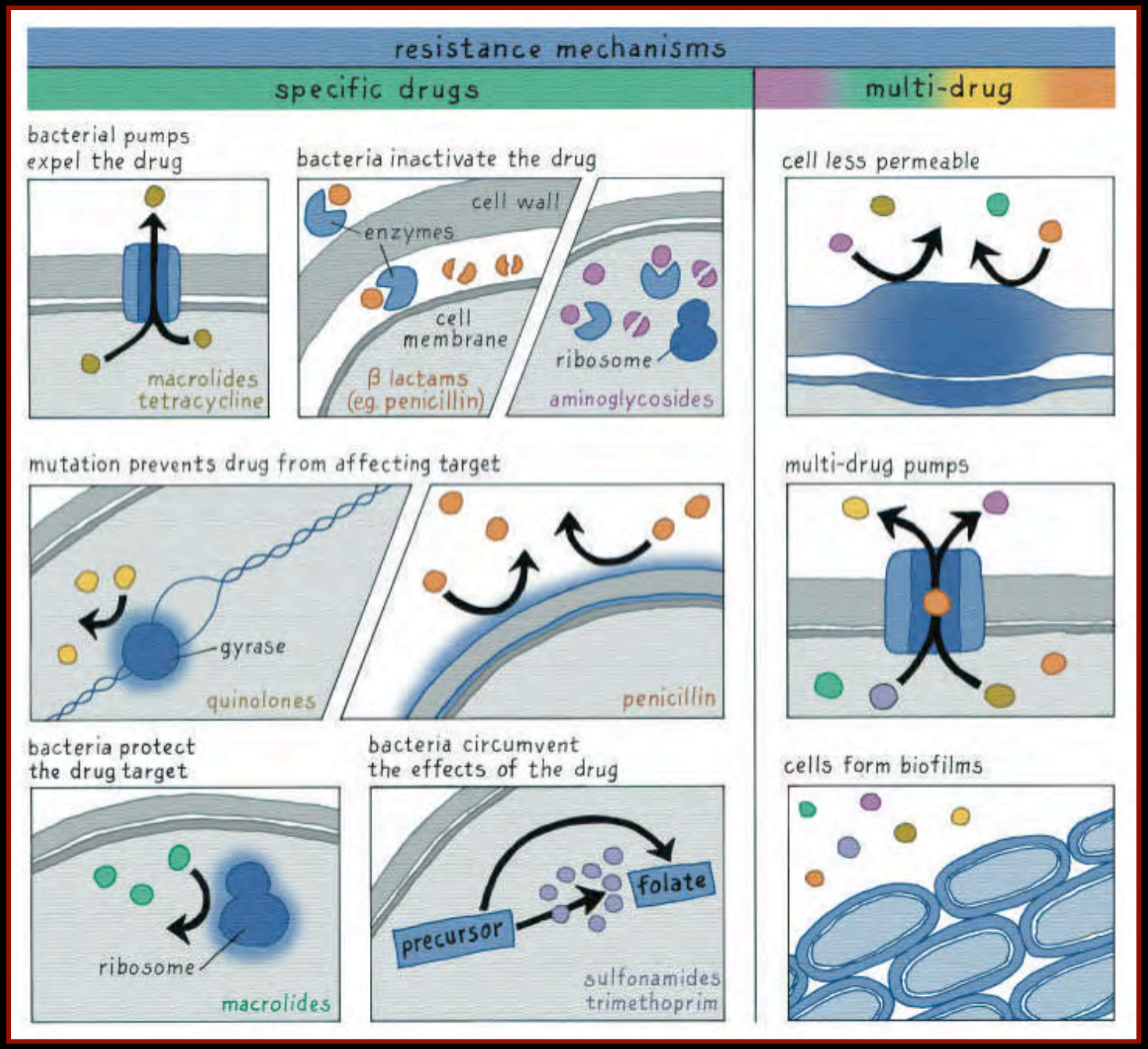




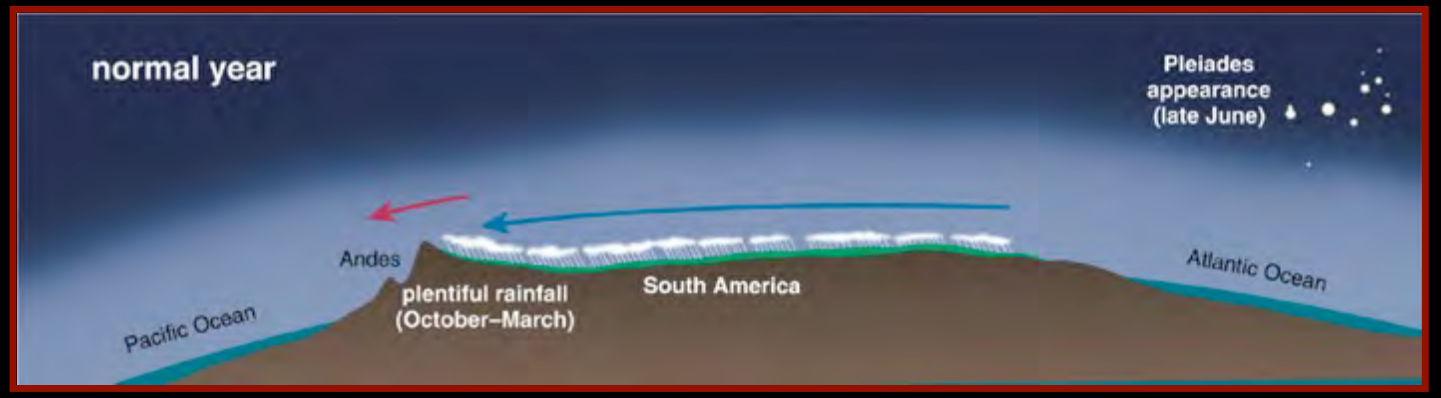
2. informal, engaging style

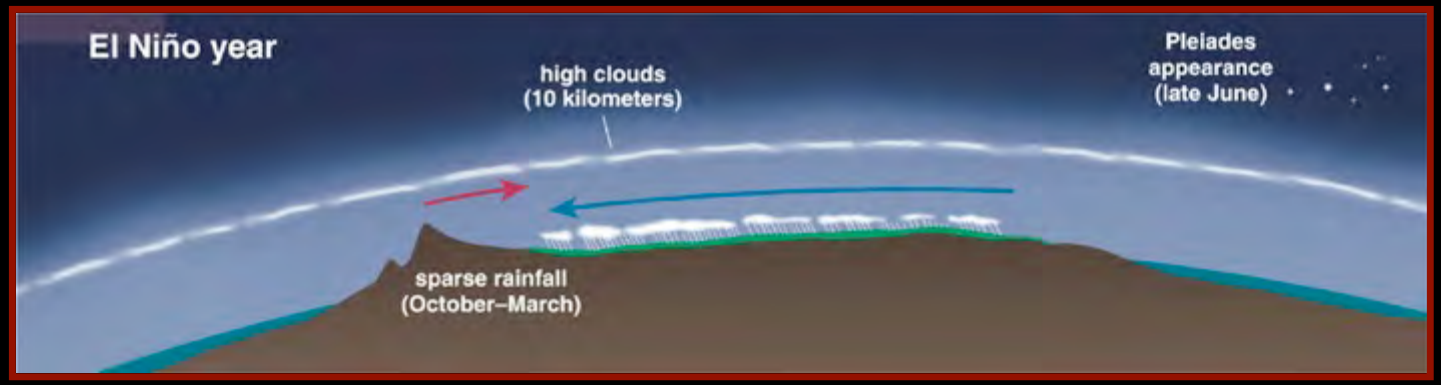

LEE

3. pictures that frame and organize arguments **as stories**


resistance mechanisms
specific drugs
multi-drug
bacterial pumps expel the drug
macrolides
tetracycline
bacteria inactivate the drug
cell wall
enzymes
cell membrane
β lactams (eg. penicillin)
ribosome
aminoglycosides
cell less permeable
mutation prevents drug from affecting target
gyrase
quinolones
penicillin
multi-drug pumps
bacteria protect the drug target
ribosome
macrolides
bacteria circumvent the effects of the drug
precursor
folate
sulfonamides
trimethoprim
cells form biofilms



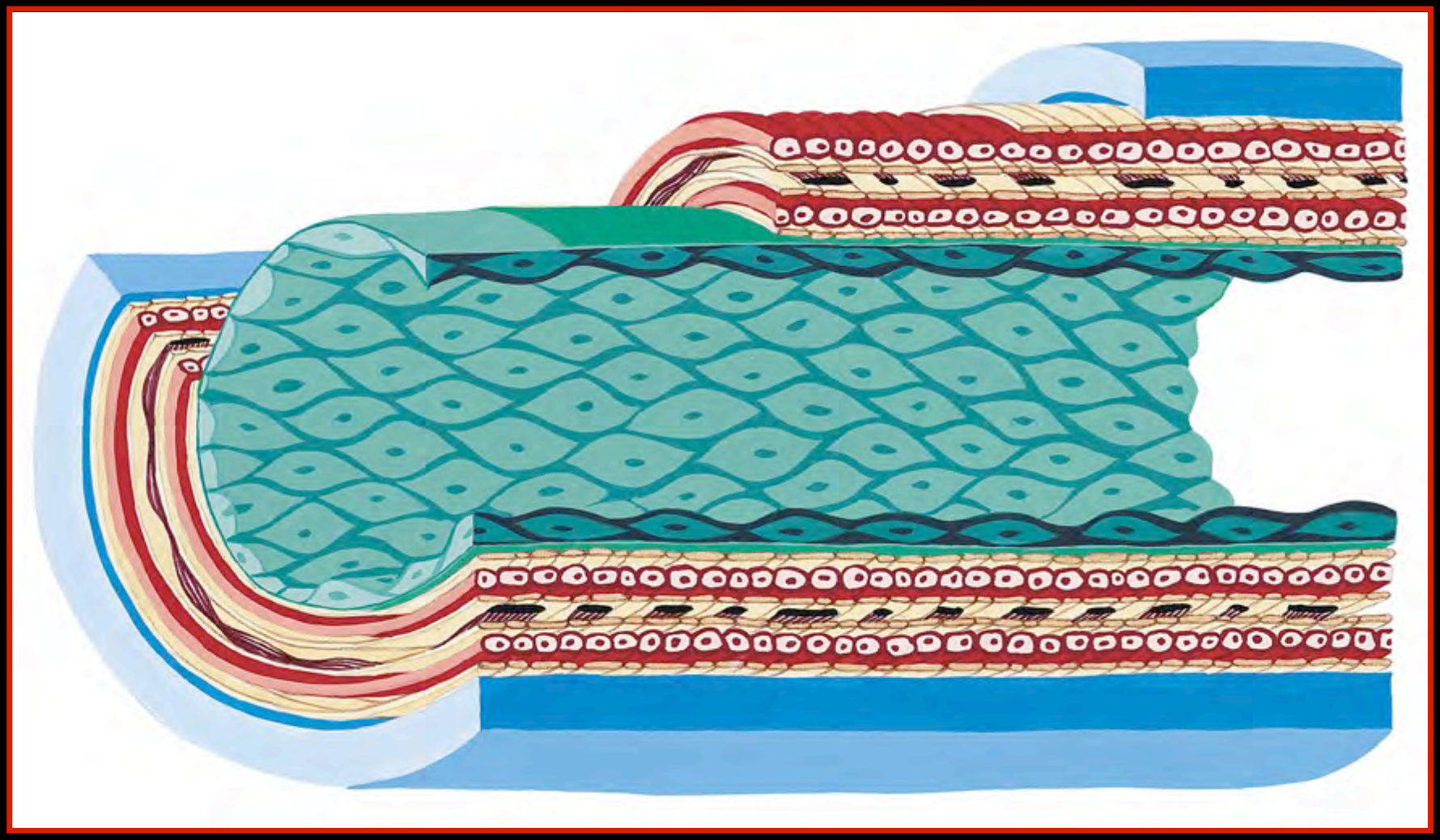

normal year
Pleiades
appearance
(late June)
Andes
Pacific Ocean
plentiful rainfall
(October–March)
South America
Atlantic Ocean

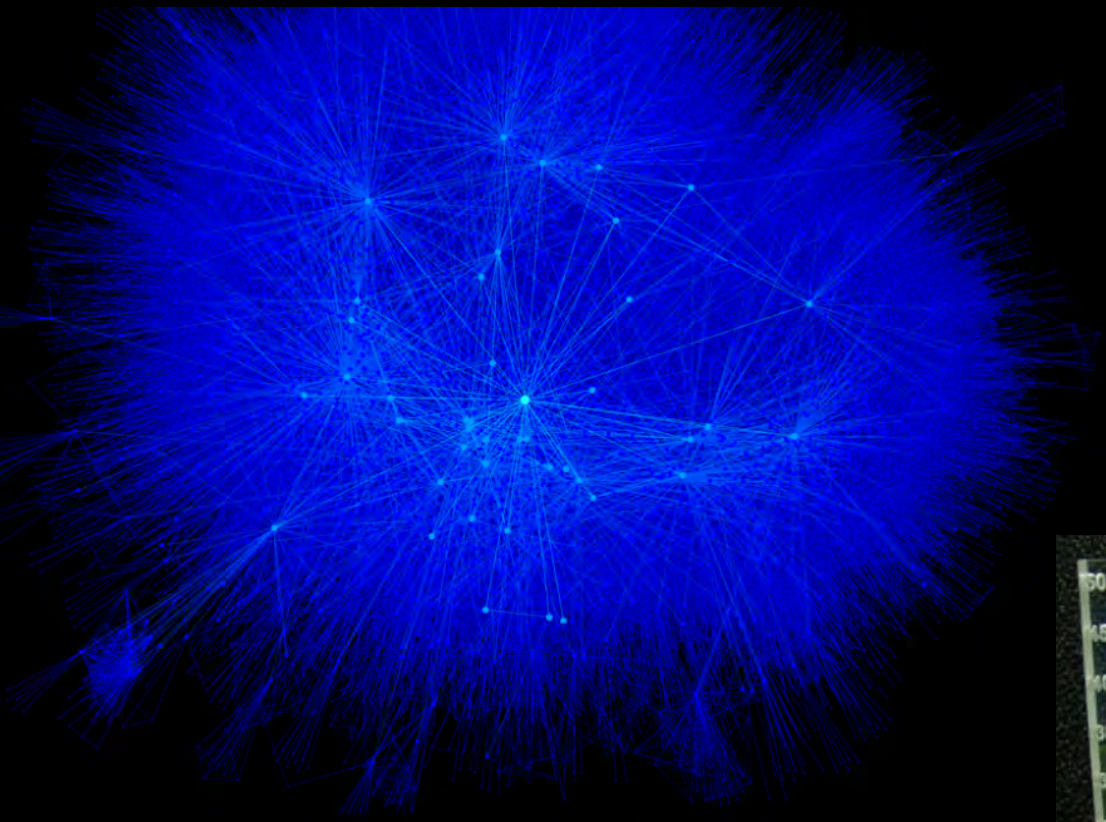

El Niño year
high clouds
(10 kilometers)
Pleiades
appearance
(late June)
sparse rainfall
(October–March)

4. simplicity


六玉河之内
紀伊高野

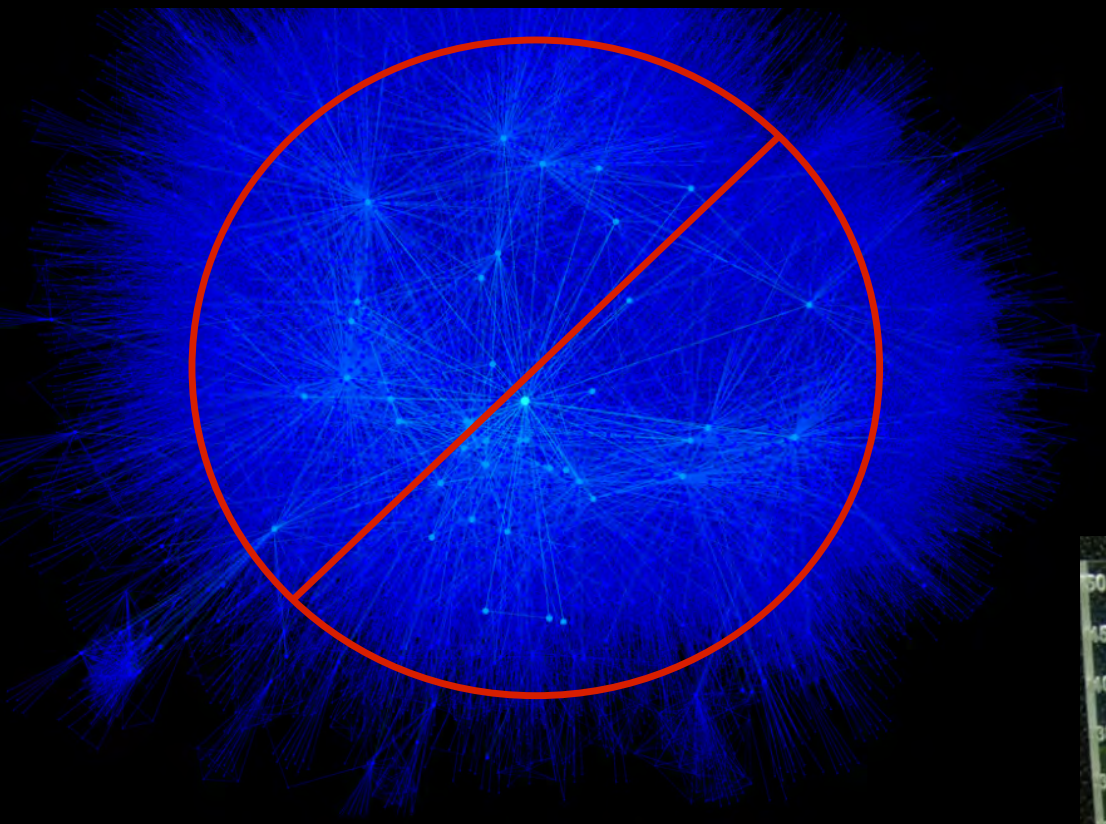

六玉河之内
陸奥野田



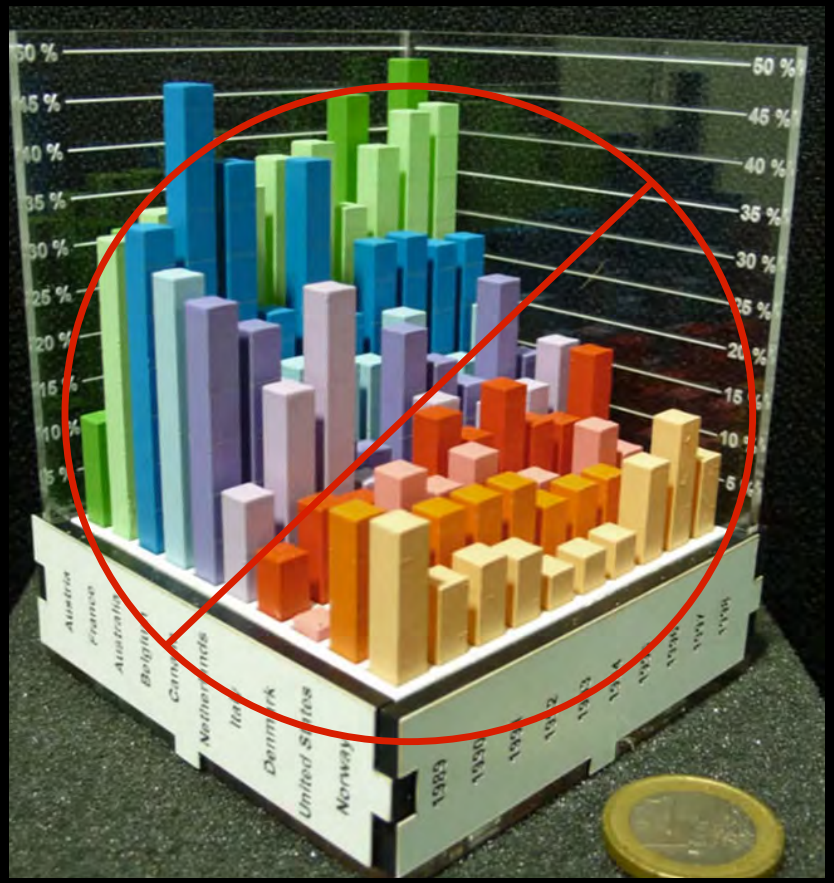


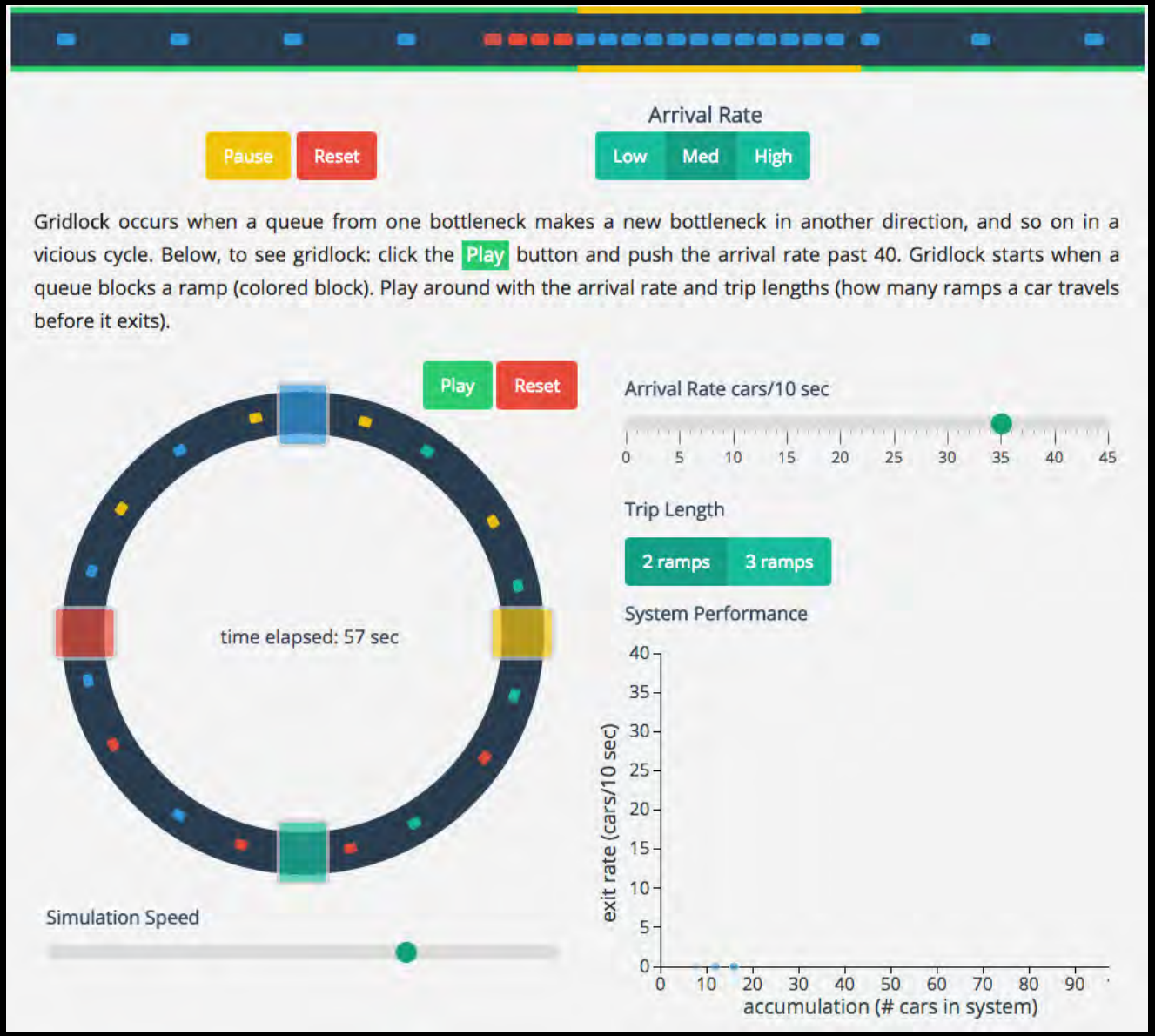
Giant Hairball



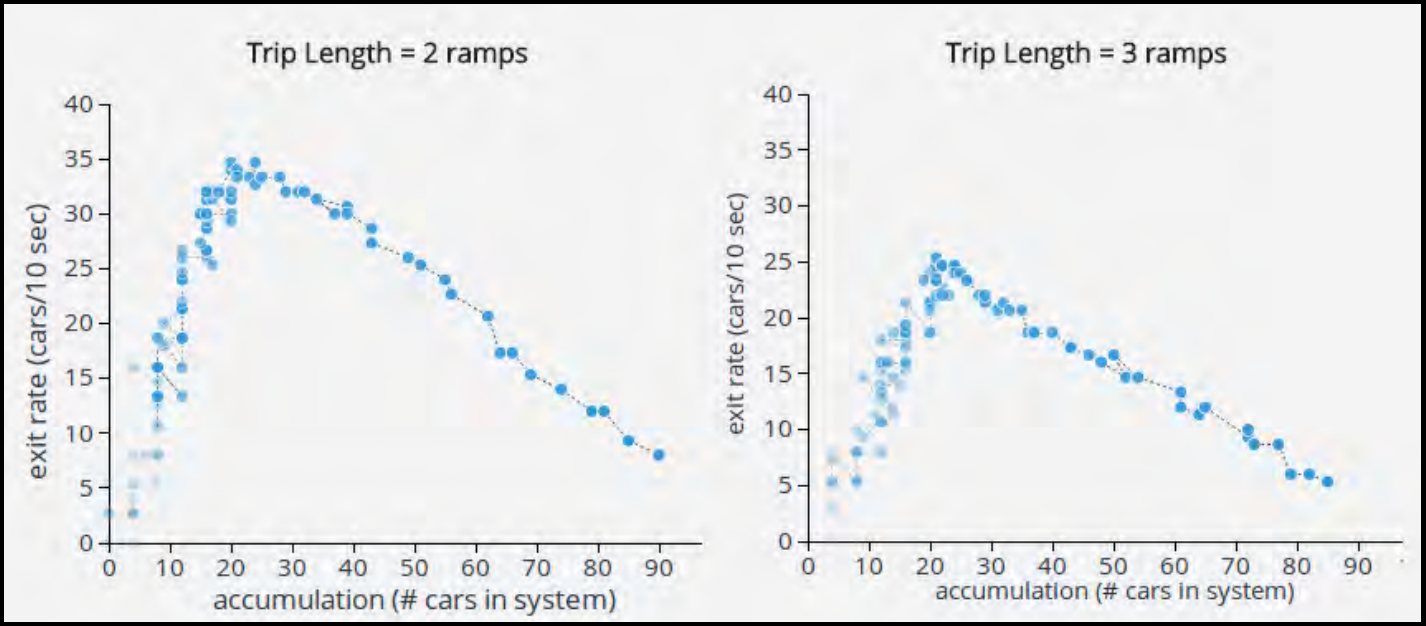


data graphics should
guide the eye to
important information





http://setosa.io/blog/2014/09/02/gridlock/



http://setosa.io/blog/2014/09/02/gridlock/

5. metaphor

San Francisco Chronicle

Violent Crime is a Beast Preying on Our City



6. humor

- conveys confidence
- adds to listener enjoyment and attention

hmm, this is a workshop.
let's do some work.

a challenge

10 min Draw a sketch that explains or demonstrates some aspect of your project.

1 min Using the sketch, briefly explain your project to the others in your group.

Tips for drawing:

- Use color to convey information.
- No jargon or symbols—use common terms for labeling.